

STRATEGY BRIEF / JULY 2026

From Parallel Agents to AI-Native Operations

How AMQ and amq-squad can support Steps 3, 4, and 5 of AI adoption

Executive thesis

AMQ should become the durable event fabric for agent communication. amq-squad should become the control plane that turns human intent into governed teams, tasks, evidence, policy, and recovery. The winning product is not more agents; it is trusted delegation with bounded autonomy.

AMQ - durable event fabric

Messages, routing, receipts, identity, wake, federation, correlation, and transport-independent delivery.

amq-squad - agent-team control plane

Goals, roles, tasks, processes, evidence, policies, human gates, lifecycle, and deterministic recovery.

The strategic boundary

Keep one source of truth per layer. AMQ owns communication and delivery semantics. amq-squad owns plans, task transitions, launch identity, policy, and operational state. This separation is what allows the system to scale without turning the message bus into an orchestrator or the orchestrator into a transport.

Source model

Based on Boris Cherny's [Steps of AI Adoption table](#) and [original post](#). This brief uses 1-based naming: Step 3 = Parallel (~10), Step 4 = Supervised autonomy (~100), and Step 5 = AI-native (~1,000+). Boris's table labels those stages 2, 3, and 4.

Prepared against amq-squad v2.22.0

THE ADOPTION MODEL

Three stages, three different products

Each stage changes the unit of management. Step 3 manages individual agents. Step 4 manages squads and lead hierarchies. Step 5 manages domain workflows and fleet policy.

Adoption step	Operating model	Primary bottleneck	Product objective
Step 3 Parallel ~10 agents	One engineer orchestrates multiple isolated agents.	Review and steering consume the engineer's attention.	Make verified output cheaper to review than agents are to supervise.
Step 4 Supervised autonomy ~100 agents	Leads coordinate child leads and specialized workers.	Trust, decision throughput, context, and exception handling.	Provide bounded autonomy with evidence, policy, and escalation.
Step 5 AI-native ~1,000+ agents	Domain workflows run continuously; humans steer by intent.	Fleet control, cost, identity, policy, and reliable automation.	Compile intent into governed workflows and surface exceptions only.

The product progression

1 Verified agent output	2 Trusted squad delegation	3 Exception-based control	4 Intent-driven workflows
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Strategic focus

amq-squad should win Step 3 before expanding. The best evidence for Step 4 is a 5-10 agent workflow that users trust enough to stop watching. The best evidence for Step 5 is real workload pressure that makes the local backend, rather than orchestration quality, the limiting factor.

STEP 3 / PARALLEL ORCHESTRATION

Make 5-10 agents boringly reliable

The objective is to make evidence cheaper to review than agents are to supervise. The human supplies the goal, approves the prepared run, and reviews exceptions plus the final verified result.

Target operating loop



Leverage AMQ

Durable task and review messages; focused threads; structured kinds; delivery receipts; cross-session/project routing; wake and presence; concise reports that link evidence instead of embedding logs.

Leverage amq-squad

Prepared goals; team roles; isolated workstreams; lead-first readiness; task claims and dependencies; dispatch and collect; activity; immutable evidence; gates; exact stop/resume; crash recovery.

Step 3 product package

One golden workflow

Pick a high-value repeated job such as issue implementation plus QA and review. Optimize the entire path before supporting every workflow.

Verification contract

Every task declares required tests, build, lint, security checks, reviewer, expected base, and immutable evidence.

Operator UX

One prepared plan, two explicit approvals, one live status surface, and exact actions for recovery. Avoid command assembly by humans.

Trust telemetry

Measure interventions, retries, rejected reviews, evidence completeness, recovery success, and accepted work per human hour.

Exit criterion for Step 3

A lead can complete a multi-agent run while the operator leaves the terminal, returns to a concise exception/final-result view, and trusts the attached evidence enough to make a decision.

STEP 4 / SUPERVISED AUTONOMY

Build a manager-of-managers control plane

At roughly 100 agents, messages can no longer behave like chat. They must be structured operational events, and humans must move from supervising execution to handling exceptions.

1. Nested squads

A verified lead creates child squads with scoped goals, budgets, capabilities, dependencies, and acceptance contracts.

2. Capability scheduler

Assign work by model, skills, repository access, cost, risk, and available capacity instead of only by named role.

3. Context packages

Construct bounded bundles of relevant code, decisions, documentation, history, constraints, and source references.

4. Verification pipelines

Completion requires task-specific evidence and independent review. Policy decides when failure blocks, retries, or escalates.

5. Policy engine

Define which actions are automatic, which require an agent reviewer, and which require a durable human gate.

6. Exception monitor

Surface blocked work, failed verification, stale ownership, budget anomalies, conflicts, and approvals - not routine progress.

AMQ evolution for Step 4

From messages to events

Add explicit schemas for task, evidence, gate, budget, incident, decision, and completion events. Carry correlation from parent goal through child workstreams.

From terminal wake to runtime adapters

Provide event-driven adapters that can resume an agent through a supported runtime API or safe pull loop. Terminal injection should be a capability, not the universal control path.

Exit criterion for Step 4

One lead manages multiple child squads, the hierarchy recovers deterministically after interruption, automated verification handles routine quality control, and the human receives only bounded exceptions.

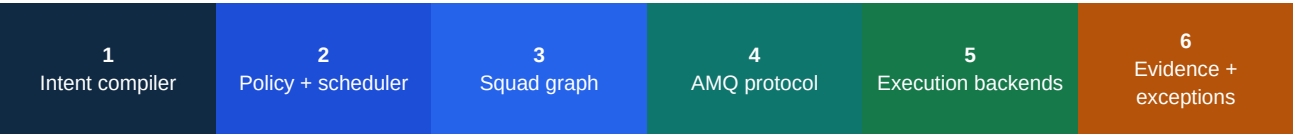
STEP 5 / AI-NATIVE OPERATIONS

Compile intent into governed fleet workflows

The local Maildir backend should remain the excellent default for development. It should not be stretched directly to thousands of agents across machines. Preserve AMQ's protocol and semantics while allowing multiple transports.

Human intent

"Upgrade 40 services to API v3. Stay under \$2,000. Do not touch production without security approval. Report only blockers and the final verified result."



Required fleet capabilities

Authenticated identity

Replace local-handle trust with authenticated agents, machines, services, and scoped authorization.

Transport abstraction

Retain the local backend while adding a distributed backend for multi-machine delivery, durability, and backpressure.

Policy and budgets

Apply organization rules for models, repositories, actions, spending, approvals, and data boundaries.

Workflow compiler

Translate domain intent into task graphs, squads, evidence requirements, gates, completion criteria, and rollback paths.

Fleet observability

Track cost, quality, latency, utilization, failure, recovery, lineage, and policy compliance with standard telemetry.

Exception routing

Route only actionable incidents to the right human or lead with complete provenance and recommended recovery actions.

Do not pre-build the fleet

Distributed infrastructure is expensive and easy to mistake for product progress. Build it only when trusted Step 4 workloads prove that single-machine transport is the active constraint.

FOUNDATION AND ROADMAP

Use v2.22.0 as the reliability base

The [v2.22.0 release](#) materially strengthens Step 3 and creates primitives needed for Step 4.

Prepared goal identity

The accepted goal, launch shape, source, digest, generation, and binding persist as one durable prepared identity.

Transactional generation

Compare-and-swap protects preparation, readiness, launch records, roster changes, goal reservation, and rollback ownership.

Claim-once delivery

The lead goal is reserved before launch and delivered once to the verified lead identity.

Lead-first readiness

Dependent workers do not launch when the configured lead is missing, mismatched, or unready.

Exact managed replay

Resume validates persisted and semantic identity before reattaching or re-orienting members.

Capability-specific terminals

Launch, visibility, input, dispatch, focus, stop, and resume are advertised only when the runtime proves support.

Priority	Outcome	Build	Proof
P0 Win Step 3	A reliable 5-10 agent workflow.	Prepared launch, task/evidence convergence, automatic review, recovery, operator UX.	Most tasks finish without corrective prompting; reviewers trust evidence.
P1 Bridge to Step 4	Trusted manager-of-managers operation.	Native runtime adapters, nested squads, capability scheduling, context packages, exception monitoring.	A lead can run multiple child squads while humans handle exceptions only.
P2 Validate Step 5	Evidence that local transport is the constraint.	Design transport abstraction and authenticated identities; avoid premature broker work.	Real workloads exceed local-machine limits and justify distributed infrastructure.
P3 Scale the fleet	Intent-to-workflow automation across an organization.	Distributed backend, policy, budgets, observability, autoscaling, domain workflow compiler.	Cost, quality, latency, and policy SLOs hold at fleet scale.

MEASUREMENT AND DECISIONS

Prove autonomy before scaling agent count

North-star metric

Accepted autonomous work per human review hour

Count work only when it meets its verification contract and is accepted without hidden cleanup. This combines throughput, quality, trust, and operator attention in one outcome metric.

Supporting metrics

Autonomy

Percent completed without corrective prompts; interventions per task; exception rate per 100 tasks.

Quality

Review rejection rate; escaped defects; evidence completeness; verification pass quality.

Reliability

Recovery success; duplicate/lost delivery rate; stale ownership; mean time to reconcile.

Economics

Cost per accepted change; token cost by workflow; fleet utilization; wasted attempts.

Primary strategic risks

Risk	Consequence	Response
Scaling before trust	More agents create a larger review and incident queue.	Gate expansion on Step 3 autonomy and quality metrics.
Terminal control dependency	Modal prompts and unsupported runtimes break safe wake-up.	Invest in native runtime adapters and explicit capability policy.
State duplication	Tasks, AMQ events, launch state, and dashboards disagree.	Keep one source of truth per layer and reconcile projections deterministically.
Infrastructure distraction	A distributed broker consumes effort before users need it.	Validate Step 5 demand with real Step 4 workload pressure.

Recommended decision

Position AMQ as the open durable event protocol and amq-squad as the local-first control plane for heterogeneous agent teams. Win the reliable 5-10 agent workflow, then expand through nested squads and runtime adapters. Treat distributed fleet infrastructure as a demand-triggered evolution, not the next feature milestone.

Sources: [Boris Cherny adoption table](#) | [Original X post](#) | [amq-squad v2.22.0](#)